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Concentration of supply chain, internal control, and corporate risk-taking

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ABSTRACT

This study uses a dataset of publicly traded companies from 2010 to 2022 to conduct a comprehensive analysis of how supply chain concentration and internal controls affect corporate risk-taking behavior. The findings reveal a significant positive relationship between supply chain concentration and the degree of corporate risk-taking. Notably, internal controls play a critical moderating role in this relationship, effectively reducing the level of risk-taking. Additionally, the research identifies distinct impacts of supply chain concentration and internal controls on risk-taking behavior between state-owned and private enterprises.

1. Introduction

Amid the wave of global economic integration, supply chain management has quietly become a core element of corporate strategy, where its efficiency and stability serves as a lifeline for enterprises, directly influencing their survival and development (Wen, Li, Sha, & Shao, 2021). As market competition intensifies and external uncertainty continues to rise, supply chain concentration—a key indicator measuring the closeness of relationships between enterprises and suppliers—has an increasingly significant impact on operational risks and the overall risk-bearing capacity of enterprises (Tian, Li, & Cheng, 2022). As such, internal control, as the foundation of a company's internal management system, plays an indispensable role in risk prevention and improving management efficiency. Therefore, a thorough analysis of the complex relationships among supply chain concentration, internal control, and corporate risk-bearing is of great importance for both academic advancement and practical decision-making in business operations.

In recent years, studies exploring the connection between supply chain concentration and corporate risk-bearing have yielded notable findings. Some scholars have argued that, although a high degree of supply chain concentration may bring advantages such as cost reduction and efficiency gains, it can also lead to supplier dependency risks, increasing operational uncertainty and financial exposure for enterprises (Zhou, Jiang, & Chen, 2022). This dependency renders enterprises particularly vulnerable in the face of emergencies such as supplier bankruptcies, supply disruptions, or price volatility. Conversely, other studies contend that close supply chain cooperation promotes information sharing and collaborative innovation, reducing transaction costs and enhancing a firm's market competitiveness. These seemingly contradictory perspectives reflect the complexity and multifaceted nature of how supply chain concentration influences corporate risk-bearing (Huang & Fan, 2022).

With respect to internal control, the academic consensus holds that a sound internal control system is essential for mitigating risk and improving management efficiency. By establishing rules and procedures, optimizing workflows, and implementing targeted actions, internal control ensures the smooth operation of corporate activities while effectively identifying, assessing, and responding to potential risks. A robust internal control system is key to reducing internal fraud, minimizing errors, and lowering a firm's overall risk level, thereby supporting stable development (Raithatha & Popli, 2022). However, most existing research examines these elements in isolation and lacks an integrated view of the interplay among supply chain concentration, internal control, and corporate risk-bearing. As a result, enterprises struggle to fully understand how these elements interact in real-world operations, which hinders the development of well-informed, scientifically grounded decisions. For this reason, constructing a comprehensive analytical framework that links supply chain concentration, internal control, and corporate risk-bearing, revealing their intrinsic connections and mechanisms—has become a focal point in current research.

The significance of this research lies in several key areas. First, by developing a comprehensive analytical framework that encompasses

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supply chain concentration, internal control, and corporate risk-bearing, this paper addresses gaps in current literature related to the interrelationship among these factors. This framework considers not only the direct effect of supply chain concentration on corporate risk-taking but also explores the crucial role of internal control in moderating this relationship. Through this structure, enterprises can gain a deeper understanding of the complex dynamics at play, enabling more informed and scientific decision-making. Additionally, this study highlights the critical function of internal control in moderating the link between supply chain concentration and corporate risk-bearing. As a central part of an enterprise's internal management system, internal control can effectively identify and manage risks through the creation of rules, optimization of processes, and implementation of specific measures, thereby reducing overall risk exposure. Under conditions of high supply chain concentration, the role of internal control becomes even more important. It enables enterprises to conduct more comprehensive assessments of potential risks (e.g., supply chain disruptions and price volatility) and to develop appropriate preventive and responsive strategies. By improving supplier management and establishing backup supply chains, enterprises can reduce the risks posed by disruptions and maintain stable development. Finally, this paper offers empirical support for government policy-making by comparing the differences in risk-bearing behavior related to supply chain management between state-owned and private enterprises. These differences are primarily reflected in areas such as resource acquisition, market positioning, and policy support. Through comparative analysis, this study reveals the different strategies and response mechanisms adopted by these two types of enterprises when managing supply chain concentration, internal control, and corporate risk-bearing. The findings offer valuable insights for government agencies in developing effective policies and helping enterprises allocate resources more rationally.

2. Theoretical analysis and research hypotheses

Supply chain concentration, as an important indicator of the degree of cooperation between enterprises and suppliers, directly affects the operational stability and risk-bearing capacity of enterprises (Marcelin, Sun, Teclezion, & Junarsin, 2022). When supply chain concentration is high, enterprises form long-term and stable partnerships with a limited number of key suppliers (Chatjuthamard, Wongboonsin, Kongsompong, & Jiraporn, 2020). This model of collaboration facilitates cost optimization, efficiency improvements, and timely information sharing, thereby supporting the sustainable and stable development of enterprises (Zhang, Shi, Yang, & Zhou, 2021). However, it also results in excessive reliance on specific suppliers (Ongsakul, Chatjuthamard, & Jiraporn, 2022). If these suppliers encounter operational difficulties such as bankruptcy, supply interruptions, or significant price fluctuations, the enterprise's normal operations can be directly affected, potentially leading to crisis (He, Song, & Chen, 2023).

To address the risks associated with such dependence, enterprises must implement a series of measures to maintain supply chain stability, such as stockpiling inventory in advance and signing long-term contracts with suppliers (Bai, Ba, Huang, & Hu, 2020). Although these measures can reduce risk to some extent, they also raise operational costs and increase the enterprise's risk-bearing capacity (Lu, Liu, & Yu, 2022). Additionally, supply chain concentration may indirectly influence an enterprise's risk-bearing capacity by affecting its market power (Li, Wang, Wu, & Zhou, 2021). In a highly concentrated supply chain system, enterprises may face significant bargaining pressure from upstream suppliers or downstream customers, resulting in shrinking profit margins (Lu, Deng, Yu, Song, & Liu, 2021). To remain competitive, enterprises may adopt more aggressive market strategies, such as increasing investment in research and development or actively expanding market share (Liu, Padgett, & Yin, 2022). However, these strategies often involve higher levels of risk, including research failures and unsuccessful market expansion. Following the theoretical analysis presented earlier, the paper introduces the first hypothesis.

H1: *The concentration of the supply chain enhances the risk-bearing capacity of enterprises.*

Internal control, as a critical internal management mechanism for enterprises, primarily aims to ensure stability, compliance, and operational efficiency, thereby significantly influencing a firm's risk-bearing capacity (Qian, Chao, & Feng, 2023). In the context of high supply chain concentration, where enterprises rely on a limited number of key suppliers, internal control plays an especially vital role (Sun, Li, Ai, & Li, 2023).

Conversely, internal control establishes a comprehensive risk management framework that enables enterprises to proactively identify, assess, and address various risks (Ye & Yue, 2023). These include risks such as supply chain disruptions, price volatility, and supplier bankruptcies (Wang, Mao, Wu, & Luo, 2023). By developing detailed risk management strategies and contingency plans, internal control can significantly reduce operational uncertainty and financial exposure caused by high supply chain concentration (Yang & Wang, 2023). For example, enterprises can adopt a diversified supplier strategy and create backup supply chains to reduce reliance on a single supplier, thereby mitigating the risks of supply chain disruption (Shang, Raza, Huo, Shahzad, & Zhao, 2023).

Conversely, internal control enhances internal management processes and strengthens the overall internal control system, thereby improving operational efficiency and risk management capabilities (Sun & Zhong, 2023). This includes better supplier management, optimized inventory control, and greater financial transparency (Lee, Li, Yu, & Zhao, 2021). By applying these measures, enterprises can respond more effectively to market competition and external changes, reducing operational costs and mitigating risks associated with high supply chain concentration.

Moreover, internal control can reduce the enterprise's risk-bearing capacity by increasing employees' risk awareness and refining their risk management skills (Lu, Song, & Yu, 2023). Training and educational initiatives can equip employees with the knowledge needed to understand the diverse risks facing the enterprise and how to respond appropriately (Lu, Zhou, Luan, & Song, 2024). This, in turn, increases employee awareness and capabilities in risk management, leading to greater attention to risk prevention and control in daily operations (Lou, Chen, Yin, Zhang, & Yu, 2022). Following the theoretical analysis presented earlier, the paper introduces the second hypothesis as follows:

H2: *Internal control reduce the risk-bearing capacity of enterprises.*

Internal control plays a significant moderating role in the relationship between supply chain concentration and an enterprise's risk-bearing capacity (Zhang, Dou, & Wang, 2023). Although supply chain concentration can offer efficiency and cost advantages, it also increases dependence on specific suppliers, thereby raising potential risks (Ye & Yue, 2024). At this point, a sound and effective internal control system provides critical support for enterprises (Chen, Yang, Zhang, & Zhou, 2020).

Internal control supports uninterrupted operations through regulation development, process optimization, and targeted implementation, while also accurately identifying, thoroughly assessing, and effectively responding to risks (Salehi, Arianpoor, & Naghshbandi, 2022). Its role becomes even more vital under conditions of high supply chain concentration (Jia, Xu, Chen, & Fernandes, 2024). It enables enterprises to analyze potential risks such as supply chain disruptions and price fluctuations in depth, thereby providing strong support for the development of preventive and responsive strategies.

By enhancing supplier management, enterprises can improve the stability and reliability of their supply networks, reducing risks associated with supplier-related issues (Qiao, He, Wang, Cai, & Zheng, 2023). Furthermore, the establishment of backup supply chains is an effective strategy for risk mitigation (Liu, Brahma, & Boateng, 2020). If the

primary supply chain encounters problems, the backup can be activated quickly, ensuring that normal operations continue without major disruptions. Following the theoretical analysis presented earlier, the paper introduces the third hypothesis as follows:

H3: *Internal control exerts a moderating influence on the link between supply chain concentration and the risk-bearing capacity of enterprises.*

Regarding the institutional environment, state-owned enterprises (SOEs) often benefit from more favorable policy support, such as tax exemptions and government subsidies, which partially reduce their risk-taking level (Ma, Ock, Wu, & Zhang, 2022). They also face fewer market entry restrictions and can more easily enter industries with high barriers to entry. In terms of resource acquisition, they typically secure low-cost financing and steady government support, enhancing their ability to withstand risks. SOEs can also access more extensive information and resources through government channels to mitigate risks associated with supply chain concentration (Andries, Balutel, Ihnatov, & Ursu, 2020). Moreover, they often hold relatively stable market positions and maintain strong relationships with suppliers and customers, which softens the impact of supply chain concentration on their risk-bearing capacity (Zhu, Yeung, & Zhou, 2021).

In contrast, private enterprises face more intense market competition and often lack sufficient resource support. When supply chain concentration is high, these enterprises may show greater dependence on specific suppliers, along with limited resources and capabilities to manage potential risks (Huang, Ma, & Wang, 2022). As a result, private enterprises face greater pressure when confronting risks such as supply chain disruptions and price volatility. However, to maintain competitiveness, they often place increased emphasis on supply chain management and optimization, which may help alleviate some of the negative effects of supply chain concentration on their risk-bearing capacity. Following the theoretical analysis presented earlier, the paper introduces the fourth hypothesis as follows:

H4: *Compared with SOEs, the impact of supply chain concentration on the risk-bearing capacity of private enterprises is more pronounced.*

For SOEs, internal control mechanisms typically emphasize compliance and stability to protect state-owned assets and implement national policy directives (Rashid, Akmal, & Shah, 2024). Although this approach may constrain flexibility and innovation, it also limits risks caused by decision-making errors or weak internal management (Si & Li, 2022). Therefore, in terms of risk-bearing, SOEs tend to adopt more conservative strategies, preferring to avoid risk through cautious operational planning.

By contrast, private enterprises tend to prioritize efficiency and flexibility in their internal controls to keep pace with rapidly changing markets and competitive pressures. This type of internal control fosters innovation and risk-taking but can also increase exposure to risks resulting from poor internal management or flawed decision-making (Connelly, Li, Shi, & Lee, 2020). As a result, private enterprises are typically more proactive in their risk-bearing behavior and more inclined to pursue rapid growth through high-risk, high-reward strategies. Following the theoretical analysis presented earlier, the paper introduces the fifth hypothesis as follows:

H5: *Compared with SOEs, the role of internal control in influencing the risk-bearing capacity of private enterprises is more pronounced/significant.*

3. Study design

3.1. Sample selection

This study focuses on publicly listed companies from 2010 to 2022 and involves the collection of relevant data that underwent rigorous screening and processing. The specific steps are as follows: first, companies classified as Special Treatment(ST), *ST, and Particular Transfer were excluded to ensure the robustness of the sample; second, samples with missing key data were removed to maintain data completeness; third, records dated before the company's public listing were eliminated to concentrate on the postlisting performance period; and finally, to minimize the influence of extreme values on the analytical results, all continuous variables were subjected to 1 % winsorization at both ends. After this series of meticulous filtering steps, a total of 34,861 valid observations were obtained.

3.2. Variable definitions

3.2.1. Dependent variable

To evaluate the level of risk-taking by enterprises, this study adopts corporate earnings volatility as the key indicator of risk ($risk-taking$). The method involves a three-year rolling window that spans the previous year ($t-1$), the current year (t), and the following year ($t+1$). During this period, the return on assets (ROA) is first adjusted based on the industry average to eliminate the effects of industry-level differences. Then, within this observation window, the adjusted ROA values serve as the basis for assessing the enterprise's risk-taking level. This approach provides a more accurate reflection of company performance under risk while preserving comparability and timeliness across different industries.

3.2.2. Independent variables

This study measures a company's supply chain concentration ($chain$) by averaging the sum of procurement ratios from the top five suppliers and sales ratios to the top five customers. By integrating both supplier and customer concentration, this metric offers a more complete depiction of supply chain risk. This approach acknowledges that risks may arise from both upstream suppliers and downstream customers. Using the average of procurement and sales ratios ensures that neither side of the supply chain is neglected, avoiding the limitations of a single-dimensional measure.

The natural logarithm of the Dibo Internal Control Index is used to assess the quality of internal control (icq). This index is built on dimensions of compliance with internal control standards, reporting accuracy, asset protection, operational efficiency, and strategic goal achievement. It also incorporates internal control deficiencies as a modifying factor, offering a comprehensive measure of a firm's internal control strength and its risk management capability. A higher icq value reflects stronger internal control quality.

3.2.3. Moderator variable

Internal Control (icq): Internal control serves not only as an explanatory variable in this study but also as a moderating variable.

3.2.4. Control variables

The control variables used in this study include firm size ($size$), debt-to-asset ratio ($lever$), years since listing (age), ownership type ($state$), fixed asset ratio ($fixed$), cash flow ratio ($cash$), revenue growth rate ($growth$), and equity balance degree ($balance$).Table 1 lists the definitions of all variables.

3.3. Model construction

To assess the effect of supply chain concentration and internal controls on corporate risk-taking, this paper establishes Models 1 and 2 for panel regression analysis:

$$risk-taking_{i,t} = \alpha_0 + \alpha_1 chain_{i,t} + \sum_{k=1}^{n} \alpha_k control_{i,t} + \psi_i + \zeta_t + \varepsilon_{i,t} \quad (1)$$

$$risk-taking_{i,t} = \eta_0 + \eta_1 icq_{i,t} + \sum_{k=1}^{n} \eta_k control_{i,t} + \psi_i + \zeta_t + \varepsilon_{i,t} \quad (2)$$

To examine the moderating effect of internal control, this paper

Table 1
Variable definitions.

	Variable Name	Variable Symbol	Variable Definition
Dependent variable	Enterprise risk-taking	*risk-taking*	Adjust the company's return on assets based on the average level of various industries to eliminate the impact of industry differences on the results. Subsequently, within a rolling time observation window, utilize this adjusted ROA value as a basis for measuring the level of enterprise risk-taking
Independent variables	Supply chain concentration	*chain*	(Ratio of purchases from the five largest suppliers + Ratio of sales to the five largest customers)/2
	Internal control	*icq*	ln(Dibo internal control index)
Moderator variable	Internal control	*icq*	ln(Dibo internal control index)
Control variables	Enterprise size	*size*	ln(total assets)
	Debt-to-asset ratio	*lever*	Total liabilities/Total assets
	Listing age	*age*	ln(current year − year of listing +1)
	Enterprise nature	*state*	A value of one is assigned to state-owned enterprises, while all others are assigned a value of 0
	Proportion of fixed assets	*fixed*	Net fixed assets/Total assets
	Cash flow ratio	*cash*	Operating cash flow to total assets ratio
	Operating revenue growth rate	*growth*	(Current year operating revenue/Previous year operating revenue) − 1
	Equity balance degree	*balance*	Ratio of shares held by the second-largest shareholder to those held by the largest shareholder

establishes Model 3 for empirical analysis:

$$risk-taking_{i,t} = \delta_0 + \delta_1 chain_{i,t} + \delta_2 icq_{i,t} + \delta_3 chain_{i,t} * icq_{i,t} + \sum_{k=1}^{n} \delta_k control_{i,t} + \psi_i + \zeta_t + \varepsilon_{i,t}. \quad (3)$$

4. Empirical analysis

4.1. Descriptive statistical analysis

The overall risk-taking capacity of enterprises (*risk − taking*) is relatively low but shows some fluctuation, indicating variations in risk tolerance across different enterprises. The concentration level of supply chains (*chain*) is moderate, though significant disparities exist, with some firms exhibiting high concentration and others being more dispersed. The quality of internal controls (*icq*) is generally high, though improvement is still possible. The average size of enterprises (*size*) is relatively large, and the debt-to-asset ratio (*lever*) is at a moderate level; however, there are substantial differences in financial conditions across firms. These enterprises also have a relatively short listing history (*age*) and an even distribution between SOEs (*state*) and non-SOEs. The fixed asset ratio (*fixed*) and the cash flow ratio (*cash*) are at moderate levels, though there is notable variation in cash flow from operating activities among enterprises. The overall growth rate of operating revenue (*growth*) is positive, reflecting a general upward trend, although the degree of growth varies across firms. Lastly, the equity balance degree (*balance*) is moderate, suggesting that some mechanisms for checks and balances exist within the ownership structures of these enterprises.

Table 2
Descriptive statistics.

	Obs	Mean	SD	Min	Median	Max
risk-taking	34,861	0.0041	0.0641	−0.2641	0.0032	0.1841
chain	34,861	0.3138	0.1687	0.0300	0.2933	0.8172
icq	34,861	6.1855	1.2991	0.0000	6.4877	6.7194
size	34,861	22.2030	1.3136	19.7810	22.0076	26.2979
lever	34,861	0.4212	0.2088	0.0545	0.4099	0.9293
age	34,861	2.1685	0.8402	0.1427	2.3192	3.3767
state	34,861	0.3846	0.4865	0.0000	0.0000	1.0000
fixed	34,861	0.2079	0.1582	0.0008	0.1756	0.6927
cash	34,861	0.0474	0.0690	−0.1609	0.0467	0.2444
growth	34,861	0.1365	0.3181	−0.5636	0.0981	1.6456
balance	34,861	0.3662	0.2859	0.0105	0.2885	0.9963

Table 2 presents the detailed descriptive statistics for each variable.

4.2. Basic regression analysis

Table 3 presents the regression results. In Column 1, supply chain concentration (*chain*) and internal control (*icq*) are used as independent variables, with corporate risk-taking (*risk − taking*) as the dependent variable, and fixed effects are applied. Column 2 includes time effects in addition to the fixed effects. Column 3 shows results after controlling for individual effects. Finally, Column 4 reports results after controlling for both individual and time effects. The results in Table 3 indicate that the double fixed effects model yields the most robust findings, with coefficients for supply chain concentration (*chain*) and internal control (*icq*) both statistically significant at the 1 % level. Therefore, **H1** and **H2** are supported. Variations in supply chain concentration significantly affect a firm's level of risk-taking. High concentration may reduce a company's bargaining power, placing it at a disadvantage in negotiations with suppliers or customers. Additionally, excessive dependence on a limited number of suppliers or customers may increase difficulties in cost control and reduce supply chain stability. These dynamics compel firms to enhance their risk management strategies and improve their ability to navigate uncertainty, which may, in turn, lead to a higher level of risk-taking.

Table 3
Basic regression results.

Variables	1	2	3	4
	risk-taking	*risk-taking*	*risk-taking*	*risk-taking*
chain	−0.0020	0.0016	0.0055**	0.0062***
	(−1.1400)	(0.8754)	(2.0060)	(3.1962)
icq	−0.0016***	−0.0015***	−0.0007***	−0.0009***
	(−6.8504)	(−6.4303)	(−2.7302)	(−2.9007)
size	0.0097***	0.0103***	0.0098***	0.0110***
	(35.4208)	(36.7482)	(15.8344)	(16.3372)
lever	−0.0896***	−0.0911***	−0.1234***	−0.1266***
	(−53.0266)	(−53.9817)	(−47.0398)	(−47.4853)
age	−0.0103***	−0.0101***	−0.0173***	−0.0150***
	(−25.5661)	(−24.8196)	(−21.9519)	(−13.1964)
state	0.0047***	0.0037***	0.0025***	0.0028***
	(7.0324)	(5.5658)	(4.3725)	(5.0148)
fixed	−0.0627***	−0.0647***	−0.0585***	−0.0616***
	(−33.1650)	(−34.0646)	(−16.1057)	(−16.8969)
cash	0.3086***	0.3095***	0.1884***	0.1839***
	(70.0934)	(69.9793)	(40.4686)	(39.4361)
growth	0.0383***	0.0390***	0.0363***	0.0381***
	(42.1465)	(42.2401)	(42.3412)	(43.6560)
balance	−0.0078***	−0.0072***	−0.0088***	−0.0081***
	(−7.6525)	(−7.0939)	(−4.7642)	(−4.4119)
Constant	−0.1662***	−0.1755***	−0.1277***	−0.1570***
	(−29.3893)	(−30.5151)	(−10.0861)	(−11.1437)
ID FE	NO	NO	YES	YES
Year FE	NO	YES	NO	YES
Observations	34,861	34,861	34,861	34,861
R^2	0.3134	0.3198	0.3045	0.3349

4.3. Moderating effect

The findings in Column 1 of Table 4 indicate that the coefficient for supply chain concentration (*chain*) is 0.0062, which is statistically significant at the 5 % level. The significance test also shows that the coefficient for internal control (*icq*) is −0.0005, confirming statistical significance and aligning with the core regression analysis on the effects of supply chain concentration and internal control on corporate risk-taking. In Column 2 of Table 4, the coefficient for the interaction term between supply chain concentration and internal control (*chain* × *icq*) is significantly negative, indicating that internal control plays a substantial moderating role in the relationship between supply chain concentration and corporate risk-taking. These results support **H3**.

4.4. Heterogeneity analysis

The regression results presented in Table 5 clearly show that for both SOEs and private enterprises, the coefficient for supply chain concentration (*chain*) trends positively. This suggests that higher supply chain concentration tends to increase the risk-taking behavior of enterprises. However, a distinct difference appears in the statistical significance of these coefficients between the two types of enterprises. Specifically, the coefficient for supply chain concentration is statistically significant only in the private enterprise sample, although it is not significant in the SOE sample. This suggests that the impact of supply chain concentration on corporate risk-taking is more pronounced for private enterprises than for SOEs. Therefore, **H4** is validated.

Drawing from the regression results shown in Table 6, it is evident that the coefficient for internal control (*icq*) is negative in both the SOEs and private enterprise samples, indicating that internal control may suppress enterprise risk-taking to some extent. However, the significance of the coefficient differs notably between the two types of enterprises. The internal control coefficient is significant only in the private enterprise sample; however, it does not reach statistical significance in the SOE sample. This suggests that the effect of internal control on risk-taking is more pronounced in private enterprises than in SOEs. Therefore, **H5** is validated. Private enterprises often display greater flexibility and efficiency in the implementation of internal systems, enabling them to respond quickly to market shifts and to adjust internal control strategies accordingly. Although their oversight frameworks may not be as formalized as those in SOEs, once internal control mechanisms are in place, they tend to have a more direct impact on operations by reducing hierarchical barriers. This efficient implementation and direct oversight enable internal controls to be more effective in lowering risk and increasing sensitivity to risk-taking within private enterprises.

Table 4
Moderating effect.

Variables	(1)	(2)
	risk-taking	*risk-taking*
chain	0.0062**	0.0128***
	(2.1962)	(2.7409)
icq	−0.0005**	−0.0008***
	(−2.39007)	(−2.7987)
Chain × *icq*		−0.0121***
		(−2.9656)
size	0.0110***	0.0110***
	(16.3372)	(16.3651)
lever	−0.1266***	−0.1266***
	(−47.4853)	(−47.4385)
age	−0.0150***	−0.0150***
	(−13.1964)	(−13.2129)
fixed	−0.0616***	−0.0614***
	(−16.8969)	(−16.8255)
cash	0.1839***	0.1840***
	(39.4361)	(39.4452)
growth	0.0381***	0.0381***
	(43.6560)	(43.5753)
balance	−0.0081***	−0.0081***
	(−4.4119)	(−4.4071)
Constant	−0.1570***	−0.1601***
	(−11.1437)	(−11.0827)
ID FE	YES	YES
Year FE	YES	YES
Observations	34,861	34,861
R_2	0.3149	0.3371

Table 5
Heterogeneity analysis (1).

Variables	State-owned enterprises	Private enterprises
	risk-taking	*risk-taking*
chain	0.0016	0.0106***
	(0.4133)	(2.5980)
size	0.0080***	0.0136***
	(8.6590)	(14.8081)
lever	−0.1273***	−0.1267***
	(−32.8257)	(−35.8545)
age	−0.0089***	−0.0173***
	(−4.3427)	(−11.4041)
fixed	−0.0479***	−0.0719***
	(−9.7423)	(−13.7524)
cash	0.1485***	0.2071***
	(21.9926)	(32.7295)
growth	0.0314***	0.0415***
	(23.3460)	(36.3987)
balance	−0.0066***	−0.0096***
	(−2.5905)	(−3.7534)
Constant	−0.0955***	−0.2125***
	(−4.6417)	(−11.0258)
ID FE	YES	YES
Year FE	YES	YES
Observations	13,408	21,453
R^2	0.3848	0.3346

Table 6
Heterogeneity analysis (2).

Variables	State-owned enterprises	Private enterprises
	risk-taking	*risk-taking*
icq	−0.0004	−0.0013***
	(−1.5577)	(−2.9338)
size	0.0077***	0.0130***
	(8.1714)	(14.1447)
lever	−0.1261***	−0.1261***
	(−31.9927)	(−35.2538)
age	−0.0089***	−0.0178***
	(−4.3636)	(−11.6881)
fixed	−0.0477***	−0.0725***
	(−9.7118)	(−13.8980)
cash	0.1482***	0.2070***
	(21.9365)	(32.6973)
growth	0.0314***	0.0416***
	(23.3205)	(36.5555)
balance	−0.0065**	−0.0096***
	(−2.5577)	(−3.7309)
Constant	−0.0912***	−0.2009***
	(−4.4815)	(−10.6339)
ID FE	YES	YES
Year FE	YES	YES
Observations	13,408	21,453
R^2	0.3850	0.3344

4.5. Endogeneity test

To address potential endogeneity concerns between supply chain concentration and corporate risk-taking, this study employs an instrumental variable approach, using lagged supply chain concentration (*L.chain*) as the instrument. The results of this test are presented in Table 7. The accompanying tests for nonidentifiability and weak instruments confirm that the selected instrumental variable is appropriate.

Table 7
Endogeneity test (1).

Variables	1	2
	Phase one	Phase two
	chain	*risk-taking*
L.chain	0.8055*** (248.1527)	
chain_pre		0.0076*** (2.8321)
Controlvariable	Yes	Yes
LM value (nonidentification test)	20,519.5037	
First stage F value (weak instrument variable test)	61,579.5524	

In the preliminary analysis, the instrument exhibits strong predictive power for current supply chain concentration. The subsequent analysis also confirms a significant positive relationship between supply chain concentration and corporate risk-taking, passing the statistical significance test. These results indicate that an increase in supply chain concentration significantly contributes to a rise in corporate risk-taking.

To further address endogeneity concerns stemming from possible sample selection bias, this study applied a propensity score matching approach. Eight covariates were selected based on the control variables, and sample matching was conducted accordingly. The one-to-three nearest neighbor matching principle was applied, with equity balance considered to ensure a highly comparable sample set. A regression analysis was then performed using the matched samples. Table 8 presents the results, showing that the coefficients for supply chain concentration and internal control remain consistent in both sign and significance when compared to the original basic regression analysis. These findings suggest that, even after accounting for potential endogeneity, the core regression results of the study remain valid and robust.

5. Conclusions

Following a thorough analysis of publicly listed companies from 2010 to 2022, this study arrived at several key conclusions. First, the findings revealed a clear positive correlation between a company's supply chain concentration and its risk-taking level, highlighting the critical role of supply chain management in corporate risk strategy. This is especially important for firms with highly concentrated supply chains, which require more careful risk management. Second, internal control was identified as a key moderating factor in the relationship between supply chain concentration and corporate risk-taking. Strong internal control mechanisms enable firms to more effectively identify, evaluate, and respond to risks associated with supply chain concentration. Moreover, internal control was shown to suppress corporate risk-taking overall. Finally, the study found differing impacts of supply chain concentration and internal control on risk-taking behavior between SOEs and private enterprises. Owing to their unique institutional status and access to resources, SOEs show greater resilience to risks associated with supply chain concentration. In contrast, private firms, facing more intense market competition, tend to rely on agile and flexible internal control mechanisms to adapt and manage risk. These findings highlight how differences in ownership structure shape the way firms approach supply chain management and internal control, ultimately influencing their risk-taking behavior.

Although this study offers an in-depth analysis of the relationships among supply chain concentration, internal control, and corporate risk-taking, it is not without limitations. First, the dataset was limited to listed companies from 2010 to 2022, which may not fully capture the conditions of nonlisted enterprises or reflect broader market dynamics. Second, to simplify model construction, some real-world complexities may have been abstracted, potentially limiting the generalizability of the findings. Future research should broaden the data scope by incorporating nonlisted firms and refine the model to better reflect real-world intricacies, thereby improving the study's accuracy and practical relevance.

Table 8
Endogeneity test (2).

Variables	*risk-taking*
chain	0.0071*** (3.4357)
icq	−0.0012*** (−2.9832)
size	0.0114*** (16.4728)
lever	−0.1245*** (−44.3792)
age	−0.0148*** (−12.7936)
state	0.0026*** (4.8247)
fixed	−0.0608*** (−16.5034)
cash	0.1827*** (39.1532)
growth	0.0379*** (43.2718)
balance	−0.0075*** (−4.3852)
Constant	−0.1570*** (−10.7042)
ID FE	YES
Year FE	YES
R^2	0.3402

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Declaration of competing interest

None.

Data availability

The authors do not have permission to share data.

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